

Solar Cell Tech Report

1. Manufacturer's name and contact information

Maxeon Solar Technologies
+1 669 3215888

2. Stock number, type, or description

Type: MAXEON™ GEN III SOLAR CELLS.
Description: These are high-performance solar electric technology cells designed and manufactured by SunPower, known for producing 25-35% more power compared to conventional cells.

3. Manufacturer's quote for cell area (cm^2)

Cell Area: Approximately **153 cm^2**.

4. Manufacturer's quote for performance

The performance of the **Maxeon Gen III Cell** at Standard Test Conditions (STC)

Characteristic	NE3 Cells	ME3 Cells
Efficiency (%)	Min 24.4	24.00 - 24.4
Power (Wp)	Min 3.78	3.72 - 3.78
Vmpp (V)	0.621 - 0.630	0.616 - 0.626
Impp (A)	6.05 - 6.11	6.01 - 6.09
Voc (V)	0.728 - 0.735	0.724 - 0.734
Isc (A)	6.39 - 6.43	6.37 - 6.42
Grade	High Aesthetic, prime cells	High Aesthetic, prime cells

Wafer	Monocrystalline silicon	Monocrystalline silicon
Design	All back contact	All back contact
Front	Uniform, black antireflection coating	Uniform, black antireflection coating
Back	Tin-coated, copper metal grids	Tin-coated, copper metal grids
Cell Area (cm ²)	Approx 155	Approx 155
Cell Weight (g)	Approx 6.6	Approx 6.6
Cell Thickness	150 micron +/- 30 micron	150 micron +/- 30 micron

5. Cell area (cm²)

39,995.9cm², 3.99m²

6. A detailed layout map of the vehicle, showing all cell types/sizes and locations, as well as calculations of total area

6.1 Total Area Calculation

One cell = 12.5cm * 12.5cm = **156.25**

$$4x1 = 156.25 * 4 = 625$$

$$4x2 = 156.25 * 8 = 1250$$

$$4x3 = 156.25 * 12 = 1875$$

$$3x1 = 156.25 * 3 = 468.25$$

$$3x3 = 156.24 * 9 = 1406.25$$

$$2x1 = 156.25 * 2 = 312.5$$

$$\begin{aligned} \text{Total Area} &= 625 * 2 + 1250 * 2 + 1875 * 13 + 468.25 * 9 + 1406.25 * 5 + 312.5 * 2 \\ &= \mathbf{39,995.9} \end{aligned}$$

6.2 Cell Physical Characteristics

Wafer: Monocrystalline silicon

Design: All back contact

Front: Uniform, black antireflection coating

Back: Tin-coated, copper metal grid

Cell Area: Approximately 153 cm²
 Cell Weight: Approximately 6.5 grams
 Cell Thickness: 150um +/- 30um

6.3 Cell Size

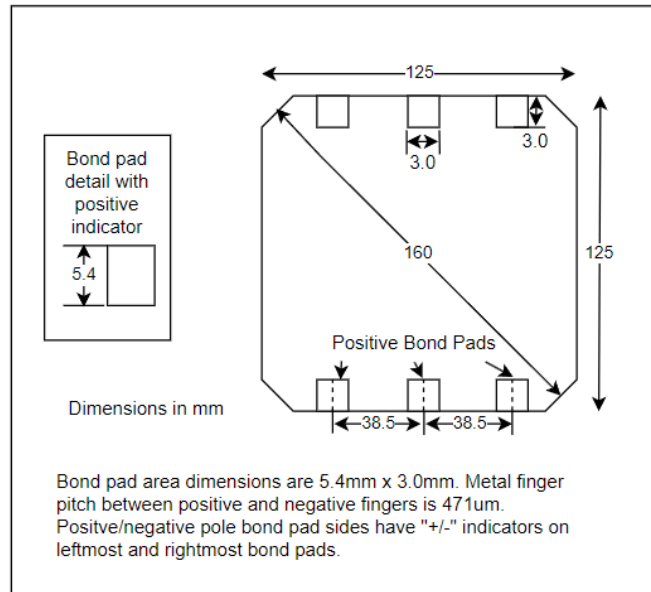


Fig 1. Cell dimensions

6.4 Detailed layout map of the vehicle

The solar cells will be laid out in four different zones. Blue-colored cells are Me3 cells and Red-colored cells are Ne3 cells. There are 256 cells in total. Our vehicle will use Ne3 cells as primary cells and Me3 cells as back up cells.

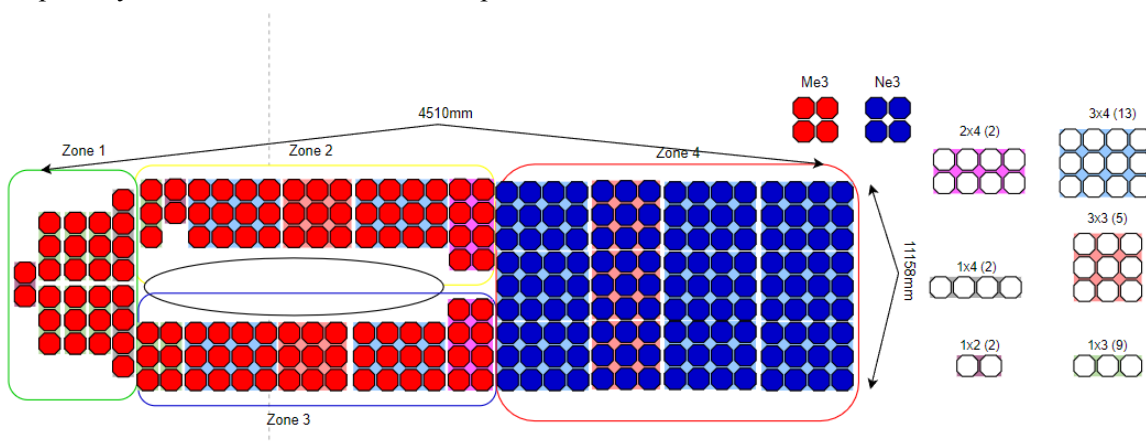


Fig 2. Solar cell layout map with array sizes, types, and locations.

Zone 1								
Module	Qty	Encap Dim (mm)	EC1	EC2	Front/Backsheet dim (mm)	BS	FS	LC Dim (mm)
4x1	2	510x130			550x150			513x131
3x1	6	383x130			420x150			387x131
2x1	1	256x130			300x150			260x131
Zone 2								
Module	Qty	Encap Dim (mm)	EC1	EC2	Front/Backsheet dim (mm)	BS	FS	LC Dim (mm)
4x3	2	510x383			550x420			513x386
4x2	1	510x256			550x300			513x260
3x3	1	383x383			420x420			387x387
3x1	1	383x130			420x150			387x131
2x1	1	256x130			300x150			260x131
Zone 3								
Module	Qty	Encap Dim (mm)	EC1	EC2	Front/Backsheet dim (mm)	BS	FS	LC Dim (mm)
4x3	2	510x383			550x420			513x386
4x2	1	510x256			550x300			513x260
3x3	1	383x383			420x420			387x387
3x1	2	383x130			420x150			387x131
Zone 4								
Module	Qty	Encap Dim (mm)	EC1	EC2	Front/Backsheet dim (mm)	BS	FS	LC Dim (mm)
4x3	9	510x383			550x420			513x386
3x3	3	383x383			420x420			387x387

Fig 3. Array types, quantity and dimensions in each zones